

11

Wk. 41 Day 284-081

OCTOBER

SATURDAY

Manish Kumar

SEPTEMBER 2025						
S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

OCTOBER 2025						
S	M	T	W	T	F	S
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Assignment.

I used Coden to help me inspect the project and by different API requests.

It helped me notice a few issues, but I checked the SQL, controller flow, and React code myself before deciding what to change.

* I focused on bugs that could show wrong task data.

Here's the bugs I found during the analysis and their best approach to resolve them:

Issue 1:-

12 Sunday

found in [TaskRepository]

Issue:-SQL search bug.

- I found this during the code base analysis manual analysis

2025

7 Lennitane

NOVEMBER 2025

DECEMBER 2025

S	M	T	W	T	F	S
						1
30						
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

S	M	T	W	T	F	S
						1
						2
						3
						4
						5
						6
						7
						8
						9
						10
						11
						12
						13
						14
						15
						16
						17
						18
						19
						20
						21
						22
						23
						24
						25
						26
						27
						28
						29
						30
						31

Day 286-079 Wk. 42

MONDAY

13

OCTOBER

and found there is an very critical issue in the SQL. that is the "SQL precedence".

As we know that the operator precedence -

() → NOT → AND → OR

but the SQL written for the business logic was wrong.

As, AND is checked before OR in SQL, and the query did not have brackets around the "title and description" condition.

Solution:- I resolved this issues by wrapping the title and description condition with parenthesis.

14

Wk. 42 Day 287-078

OCTOBER

TUESDAY

SEPTEMBER 2025

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

OCTOBER 2025

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

9 | Issue 2 :-

10 | Found in [TaskController]

11 | $\Rightarrow$ Unnecessary delay and bad request
12 | handling.1 | - While going through the request
2 | flow I noticed that
3 | there is Thread.sleep in the
4 | Controller.4 | It was deliberately waiting for during
5 | every request, which blocks a
6 | server thread and makes the
7 | search feel slow.8 | Solution:-

9 | I removed that delay.

10 | I also checked bad status, page,
11 | and page-size values. They were
12 | causing 500 errors even though
13 | they are invalid client input.14 | I added validation so that API
2025 | returns a clear 400 response instead.

NOVEMBER 2025

S	M	T	W	T	F	S
						1
30						
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

DECEMBER 2025

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Day 288-077 Wk.42

WEDNESDAY

15

OCTOBER

9 | Issue 3:-

found in [TaskController]

10 |

Issue $\Rightarrow$ Database pagination

11 |

12 |

- I noticed that the API was first loading every matching data task into a list and then creating a sublist to return one page.

2 |

3 |

4 |

5 |

this is good when the data is too small but it will use more database work, memory and network transfer as the task table gets bigger.

6 |

Solution:-

I moved pagination to the database using

Pageable/PageRequest

and a count query.

Now the database returns only 10025

16

Wk. 42 Day 289-076

OCTOBER

THURSDAY

SEPTEMBER 2025

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

OCTOBER 2025

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

9 | rows need for that page.

10 | * I kept a stable created at and id
ordering so pages do not jump around
11 | when timestamps are the same.

12 | Issue 4 :-

Found in [hook : useTask.js]

Issue :- franted Request issue.

4 | I checked what happens when a
request fails. The code set on error,
5 | but it did not turn loading off.

6 | Since the UI checks loading first,
the user could keep seeing "Loading
tasks" instead of the error.

Solution :-

I made sure loading
is cleared on both success &
failure. using finally block.

S	M	T	W	T	F	S
30					1	
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Issue 5:-

Found in [App. 01]

⇒ Resetting the page after filtering

- I tried going to a later page and then applying a narrow filter.

The app stayed on the old page no., even when that page no. no longer existed, so it showed no tasks.

Solution:-

I reset the page to page 1 when ever the search term or status changes. This is a small change, but makes filtering behave the way user expects.

Note:-

I have used Coder to also do document NOTE. md. and took help to structure my handwritten note well, along with to code as per my requirement.